

integrated demand-supply systems are essential for the Indian agricultural economy to grow. All this and more are achievable by what Intel's CEO Pat Gelsinger calls the five superpowers—ubiquitous computing, pervasive connectivity, cloud-to-edge infrastructure, sensing, and artificial intelligence. The implementation of these technologies has the potential to enable Indian farmers to adopt more sustainable and cost-effective farming practices, while also facilitating climate-smart agriculture. To accelerate the adoption of smart farming practices, we need to build Digital Readiness amongst farmers and the aligned ecosystem, enabling them to learn new tech skills, trust new technologies, and enhance the ability to use these superpowers responsibly and effectively for socio-economic benefits," says Shweta Khurana, Senior Director - APJ, Government Partnerships & Initiatives, Global Government Affairs, Intel.

However, despite these benefits, the adoption at the grassroots level is still low. Rajesh Jalan, Chief Technology Officer, Cropin, says, "India is moving ahead with the adoption of smart farming technologies. Sensors and Internet of Things (IoT), automated irrigation, satellite imaging, and drone monitoring are already being used by sections of the agricultural sector—specifically corporate-supported farms, co-operatives, and large-holder farmers. However, compared to the size of India's farm sector and the number of farms and farmers, the level of adoption is still low."

Transitioning from mere automation to smartness

Agriculture still sustains on traditional wisdom—and there are no second thoughts about that. However, it looks like farmers could do with a little technological help to deal with the changing (or challenging) environmental and market forc-



TartanSense robots help farmers with tasks like weeding, picking, and spraying (Courtesy: TartanSense)

es. Automation, with a good dose of intelligence, all strung together as a smart farming platform, could assist a farmer—and consequently the society—in several ways.

Machines. Robots, drones, and other machinery help automate several processes on a farm. They increase control over processes, reduce errors, improve productivity, and decrease wastage. Robots are used for weeding, planting seeds, harvesting agricultural produce, palletising, and more.

Take the case of fruit harvesting robots—they use sensors to detect the stage of development of each fruit. Then, based on pre-programmed parameters, they pluck only those fruits that are at the right stage of ripeness. They are also made to handle delicate crops with care.

Similarly, smart sprinklers are often used to water crops. They usually work together with sensors that measure moisture levels of air and soil. They compare sensor readings against data about the moisture requirements of the crop, and provide just the right amount of water.

Drones are also handy helpers. They help in tasks like spraying fertilisers, monitoring fields and livestock, and mapping farms. Although it seems like automation will steal people's jobs, in the agricultural sector that is not the case—there is no holiday on a farm, there is work to be done every day. But there is

comparatively less labour available in villages today, with more and more people migrating to the cities in search of work.

Sensors. Sensors are a critical aspect of climate-smart farming, enabling several tasks ranging from automated irrigation to soil testing. Sensors are used to measure—and thereby control several aspects of a farm ranging from soil quality, light, and humidity, to movement and health of livestock, and much more. Sensors are indispensable in the deployment of smart greenhouses, where crops are grown under controlled environments.

Internet of Things (IoT). The IoT is the networking fabric that brings together the robots, drones, sprinklers, sensors, actuators, and other hardware on the farm. It integrates physical and digital resources, enabling remote monitoring and management of farm activities.

For example, data from sensors may be used to initiate sprinklers wherever moisture content is low, or alert the concerned personnel when this does not happen as programmed. The IoT feeds data from varied in-farm hardware to smart software platforms.

Artificial intelligence (AI). The use of AI marks the shift from mere automation to smart farming. AI turns the data collected by the IoT into actionable insights for farmers.

For example, AI tools can study